POKHARA UNIVERSITY

Semester: Spring

Level: Bachelor
Programme: BE
Course: Advanced Programming with Java

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define the term Architectural Neutral. Write down the code to handle user defined exception for LowBalance when balance is less then 10 rupees on your mobile. 7

   b) Define multithreading. Explain the techniques to create thread using anonymous class. 8

OR

Write down the advantage of ORM over JDBC. Explain Singleton Pattern and Factory method pattern in detail.

2. a) Explain the concept of inheritance in Java. What are the benefits of using inheritance, and how does it promote code reusability and polymorphism? Provide an example to support your explanation. 8

   b) Differentiate between abstract class and interface? Explain final classes and methods. 7

3. a) Write down the JavaFx code to design a login page as shown in the figure below when user click on login button test the username is "PU" and password is "PU@2025Spring" or not. 8



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Why do we need layout management? Explain GridBag layout with suitable example. 7

a) What is the significance of stub and skeleton in RMI? Create a RMI application such that a client sends an integer number to the server and the server return the factorial value of that integer. 8

b) Create a TCP client server application where the client sends a string and the server responds by echoing the same string in upper case. 7

a) A table student consists of id, name and program. Write a program that asks users to enter a program name and displays the total number of students enrolled in that program. 7

OR

What is SQL Injection Attack? Write sample program using prepared statement to prevent SQL Injection Attack.

b) What is package in java? How can you create user–defined package in java? Explain with example. 8

6. a) Differentiate between servlet and JSP. Create a HTML file with two text fields to First name and Last name. Then create a JSP file that reads values from the HTML form and display full name. 8

b) What is cookie? How can you store and read cookies by using servlet. Explain with proper example. 7

7. Write short notes on: (Any two) 2×5

a) Hibernate ORM

b) CORBA

c) SQL Escapes

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